

LATACUNGA, Ecuador

WMO: 841230

Lat: 0.907S

Lon: 78.616W

Elev: 2806

StdP: 71.86

Time Zone: -5.00 (W05)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	4.4	0.8	5.6	9.5	2.2	6.2	9.3	11.3	16.5	10.5	16.5	1.2	10	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.2	22.9	12.9	21.9	13.2	21.0	13.0	15.0	19.1	14.4	18.8	14.0	18.6	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	14.2	15.4	13.2	13.4	14.9	12.9	13.1	14.8	53.1	18.6	51.4	19.1	50.2	18.1	18.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	9.4	8.8	-0.4	24.3	1.3	0.7	-1.3	24.9	-2.0	25.3	-2.7	25.7	-3.7	26.3
			-0.9	16.2	1.4	1.0	-1.9	17.0	-2.7	17.5	-3.5	18.1	-4.5	18.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	14.4	14.5	14.2	14.1	13.9	13.3	12.8	12.6	12.8	14.1	14.3	14.3
	DBStd	1.49	1.27	1.28	1.17	1.27	1.26	1.11	1.30	1.37	1.52	1.53	1.42	1.33
	HDD10.0	3	0	0	0	0	0	0	1	1	1	1	0	0
	HDD18.3	1660	121	107	127	127	137	152	173	177	164	132	119	125
	CDD10.0	1384	138	127	132	123	121	98	86	82	87	127	131	133
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	7	1	1	0	0	0	0	0	0	0	1	3	1
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.8	3.7	3.5	3.2	3.2	3.4	4.4	5.0	4.9	4.5	3.4	3.0	3.6
Precipitation	PrecAvg	724	62	72	90	93	64	46	26	24	46	67	73	66
	PrecMax	928	156	124	137	157	147	87	46	52	113	134	133	108
	PrecMin	588	15	24	45	27	23	18	12	13	18	26	29	21
	PrecStd	103	34	25	24	30	29	18	9	10	21	23	26	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	23.1	22.1	22.0	22.1	20.8	20.9	21.2	22.0	23.1	23.8	23.0
		MCWB	13.2	13.6	13.8	13.9	13.1	12.9	12.2	11.9	12.6	12.5	12.4	13.5
	2%	DB	21.9	21.5	20.9	20.2	20.4	19.1	19.2	19.9	20.4	22.0	22.2	21.3
		MCWB	13.3	13.5	13.3	13.4	13.1	12.6	11.9	11.6	11.9	12.9	12.8	13.2
	5%	DB	20.5	20.1	19.5	19.1	19.1	18.1	18.2	18.4	19.2	20.8	21.0	20.2
		MCWB	13.0	13.1	13.2	13.3	13.1	12.5	11.7	11.5	11.8	12.7	13.1	12.9
	10%	DB	19.0	18.9	18.2	18.0	18.1	17.1	17.1	17.2	18.1	19.1	19.2	18.9
		MCWB	12.8	12.9	12.9	12.9	12.8	12.0	11.4	11.2	11.6	12.5	12.7	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.3	15.2	15.2	14.7	14.1	13.6	13.4	14.4	14.9	15.2	15.2
		MCDB	19.9	19.8	18.0	18.2	18.7	18.0	18.4	18.4	18.5	19.0	20.2	20.1
	2%	WB	14.2	14.4	14.5	14.5	14.0	13.5	12.8	12.7	13.1	13.9	14.2	14.1
		MCDB	19.2	19.0	17.9	18.1	18.3	17.4	17.4	17.4	18.0	19.2	19.3	19.2
	5%	WB	13.7	13.7	13.9	13.9	13.6	13.0	12.3	12.1	12.5	13.3	13.7	13.6
		MCDB	18.5	18.2	17.7	17.8	17.8	16.9	16.8	16.8	17.4	18.5	18.6	18.4
	10%	WB	13.2	13.3	13.3	13.4	13.1	12.5	11.7	11.7	12.0	12.7	13.2	13.0
		MCDB	17.7	17.6	17.3	17.1	17.3	16.3	16.0	16.0	16.8	17.6	17.9	17.6
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.2	9.8	9.6	9.6	9.0	10.0	10.7	11.9	11.8	12.2	11.1
		MCDBR	13.1	12.4	11.5	10.9	11.2	10.4	11.7	12.7	14.0	14.1	14.9	13.2
		MCWBR	5.5	5.4	5.3	5.1	5.2	4.9	5.5	5.9	6.8	6.3	6.8	6.0
	5% WB	MCDBR	11.1	10.7	10.1	9.8	9.9	9.0	10.2	11.1	12.1	12.0	12.2	11.6
MCWBR		5.0	4.9	5.2	5.0	4.9	4.7	5.0	5.4	6.1	5.7	5.9	5.5	
Clear-Sky Solar Irradiance	taub	0.325	0.349	0.336	0.331	0.313	0.295	0.297	0.332	0.351	0.353	0.351	0.343	
	taud	2.539	2.457	2.513	2.535	2.593	2.660	2.638	2.479	2.416	2.435	2.446	2.480	
	Ebn at Noon	1006	982	985	968	962	968	971	954	955	966	972	983	
	Edh at Noon	108	119	112	106	96	88	91	111	121	120	117	113	
All-Sky Solar Radiation	RadAvg	3.78	3.54	3.50	3.65	3.58	3.58	3.78	4.00	4.11	4.10	4.02	3.73	
	RadStd	0.41	0.29	0.22	0.15	0.25	0.23	0.27	0.31	0.31	0.35	0.41	0.24	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page